

ALERT AP, NU, Canada

WMO: 710820

Lat: 82.520N

Lon: 62.280W

Elev: 100

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-45.7	-43.0	-48.4	0.3	-41.8	-47.4	0.3	-41.0	42.1	-6.6	36.4	-8.0	2.0	300	1.166

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	53.3	45.3	48.4	42.4	44.0	39.5	45.6	52.7	42.6	48.3	39.8	43.7	20.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
39.3	35.5	47.6	37.3	32.8	44.8	35.7	30.8	40.8	18.0	53.1	16.5	48.4	15.2	43.9	53.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.7	28.0	22.8	-49.4	59.7	3.8	3.2	-52.1	62.0	-54.3	63.9	-56.5	65.6	-59.2	67.9
			-48.0	49.0	1.9	3.1	-49.4	51.2	-50.5	53.0	-51.6	54.8	-53.0	57.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.5	-25.1	-27.8	-26.8	-11.6	10.9	30.6	37.7	33.1	16.9	-0.7	-12.8	-20.9
	DBStd	25.28	9.59	9.58	10.07	11.21	9.00	5.84	5.15	6.16	9.13	9.53	9.94	9.84
	HDD50	18082	2327	2180	2381	1847	1211	581	384	525	993	1573	1883	2197
	HDD65	23556	2792	2600	2846	2297	1676	1031	848	990	1443	2038	2333	2662
	CDD50	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	5.9	5.9	5.4	4.0	5.0	5.5	6.1	7.6	7.1	7.0	6.2	5.7	5.6
	PrecAvg	6.8	0.3	0.2	0.3	0.4	0.5	0.5	1.1	1.0	1.1	0.6	0.4	0.3
(15) Precipitation	PrecMax	11.5	1.3	0.9	1.1	1.5	1.1	1.9	2.2	2.3	2.3	1.7	1.0	0.9
	PrecMin	3.7	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.2	0.0	0.1
	PrecStd	1.4	0.2	0.2	0.2	0.4	0.3	0.4	0.6	0.6	0.6	0.4	0.3	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	10.1	8.9	23.3	35.9	51.6	57.9	55.9	41.1	27.9	19.4	10.7
		MCWB	3.0	9.6	8.0	20.9	32.8	44.4	48.6	46.8	35.8	24.6	17.3	9.1
	2%	DB	-2.2	-4.2	1.5	16.4	30.3	44.9	55.0	49.0	35.9	21.6	14.2	6.4
		MCWB	-3.4	-5.2	0.9	15.2	27.8	39.6	46.5	42.3	32.2	19.4	12.9	5.4
	5%	DB	-6.5	-9.1	-6.3	10.3	26.6	41.2	52.3	45.2	31.3	15.8	8.0	0.7
		MCWB	-7.3	-9.8	-7.0	9.5	25.1	37.7	44.7	39.9	29.2	14.5	6.8	-0.3
	10%	DB	-9.9	-14.0	-11.8	5.0	23.0	38.2	47.2	41.8	28.8	12.7	2.3	-5.1
		MCWB	-10.6	-14.6	-12.5	4.1	21.7	35.9	42.1	37.4	27.3	11.7	1.4	-5.7
	0.4%	WB	3.3	9.8	8.5	21.3	33.2	44.6	49.5	47.4	35.9	25.2	17.8	9.2
		MCDB	4.1	9.9	9.3	23.6	35.6	50.1	57.0	53.6	40.2	27.7	19.2	10.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.6	-5.1	1.4	15.5	28.2	40.1	47.1	42.6	32.4	19.6	12.9	5.6
		MCDB	-1.6	-4.5	2.2	16.6	30.1	44.4	54.8	48.7	35.3	21.6	14.4	6.7
	5%	WB	-7.4	-9.6	-6.4	9.7	25.0	37.8	44.8	40.1	29.5	14.5	7.0	0.1
		MCDB	-6.8	-9.1	-5.9	10.5	26.3	40.9	51.8	44.9	31.1	15.5	8.1	1.1
	10%	WB	-11.2	-14.4	-11.6	4.4	21.9	35.8	42.2	37.6	27.3	11.8	1.6	-5.6
		MCDB	-10.6	-13.9	-10.9	5.4	22.9	38.1	47.4	41.0	28.8	12.8	2.2	-4.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.5	12.4	12.0	8.7	6.5	8.5	7.8	9.3	11.8	11.9	11.9
		MCDBR	16.8	19.6	21.2	17.1	12.3	11.7	18.5	14.9	10.3	15.8	17.4	18.7
		MCWBR	17.7	19.4	20.7	16.2	10.6	7.9	11.8	10.0	8.2	14.4	16.8	18.5
	5% WB	MCDBR	18.4	19.6	22.0	17.0	11.4	11.4	17.8	14.9	10.0	15.5	17.6	19.1
		MCWBR	17.8	19.3	21.4	16.2	9.9	7.8	11.6	10.1	8.1	14.2	17.0	18.7
(40) Clear-Sky Solar Irradiance	taub	N/A	N/A	0.174	0.252	0.274	0.300	0.291	0.269	0.194	N/A	N/A	N/A	N/A
	taud	N/A	N/A	1.972	2.118	2.171	2.227	2.390	2.403	2.065	N/A	N/A	N/A	N/A
	Ebn at Noon	0	0	160	233	258	256	255	235	154	0	0	0	0
	Edh at Noon	0	0	10	23	29	30	24	18	10	0	0	0	0
(44) All-Sky Solar Radiation	RadAvg	0	0	163	1082	1992	2273	1913	1059	303	0	0	0	0
	RadStd	0	0	15	60	112	108	136	108	28	0	0	0	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page